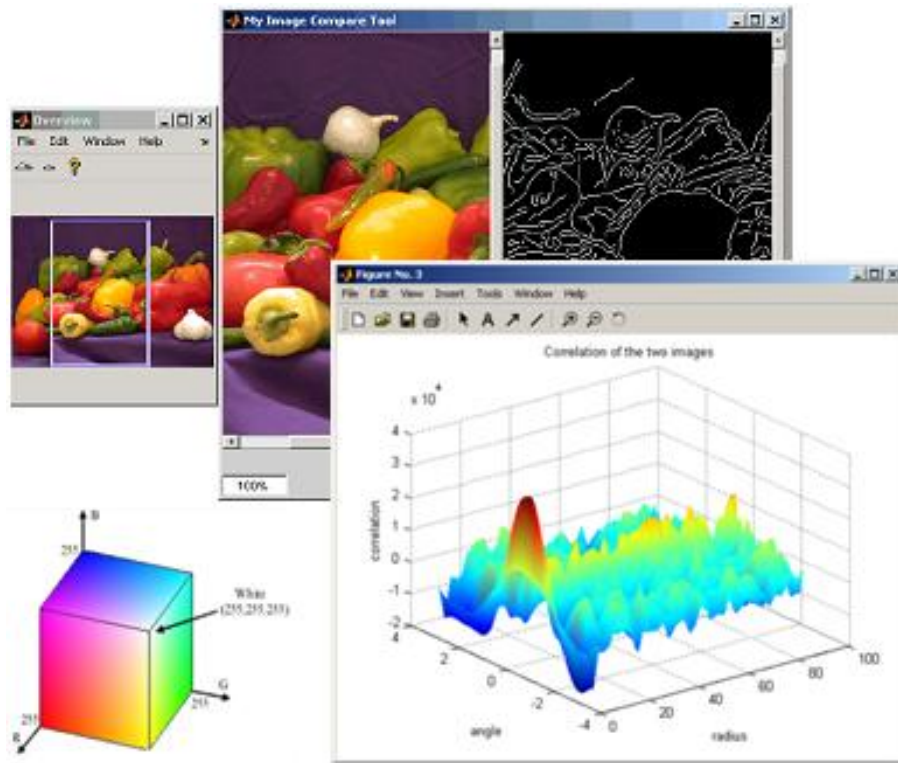


Digital Image Processing Lab

MATLAB COMMANDS FOR IMAGE PROCESSING



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Basic MATLAB Commands For Image Processing Applications

MATLAB Commands

❑ Image Loading and Saving

- **imread:** Reads an image file into MATLAB

Example: `img = imread('image.jpg');`

- **imwrite:** Writes an image matrix to a file

Example: `imwrite(img, 'output.png');`

❑ Image Display and Visualization

- **imshow:** Displays an image in a figure window

Example: `imshow(img);`

MATLAB Commands

❑ Image Conversion

- **rgb2gray:** Converts an RGB image to grayscale

Example: `grayImage = rgb2gray(img);`

- **gray2rgb:** Converts a grayscale image to RGB

Example: `rgbImage = gray2rgb(grayImage);`

- **im2double:** Converts an image to double precision

Example: `imgDouble = im2double(img);`

- **im2uint8, im2uint16:** Converts an image to uint8 or uint16 format.

MATLAB Commands

❑ Filtering and Convolution

- **imfilter**: Applies a filter to an image using convolution

Example: `filteredImg = imfilter(img, kernel);`

- **conv2**: 2D convolution of images

Example: (e.g., `convImg = conv2(img, kernel, 'same');`)

MATLAB Commands

❑ Morphological Operations

- **imdilate:** Dilates an image using a structuring element.
- **imerode:** Erodes an image using a structuring element.
- **imopen:** Performs morphological opening (erosion followed by dilation).
- **imclose:** Performs morphological closing (dilation followed by erosion).

MATLAB Commands

❑ Image Transformation

- **imresize:** Resizes an image

Example: `resizedImage = imresize(img, [256 256]);`

- **imrotate:** Rotates an image

Example: (e.g., `rotatedImage = imrotate(img, 45);`)

- **imcrop:** Crops an image

Example: `croppedImg = imcrop(img, [x y width height]);`

- **imtranslate:** Translates (shifts) an image by specified amounts.

- **imwarp:** Applies a geometric transformation to an image.

MATLAB Commands

❑ Histogram and Intensity Adjustments

- **imhist:** Displays the histogram of an image

Example: (e.g., `imhist(img);`

- **histeq:** Applies histogram equalization to enhance contrast

Example: `equalizedImg = histeq(img);`

- **imadjust:** Adjusts the intensity values of an image



Thank You